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6. (5)

$$a. \because L(A) \subseteq L(A^*), L(B) \subseteq L(B^*)$$

$$\therefore L(A \cup B) \subseteq L(A^* \cup B^*)$$

$$\because \text{若 } L_1 \subseteq L_2, \text{ 则有 } L_1^* \subseteq L_2^*$$

$$\therefore L((A \cup B)^*) \subseteq L((A^* \cup B^*)^*)$$

$$\therefore L(A) \subseteq L(A \cup B)$$

$$\therefore L(A^*) \subseteq L((A \cup B)^*)$$

$$\text{同理 } L(B) \subseteq L((A \cup B)^*)$$

$$\text{合并有 } L(A^* \cup B^*) \subseteq L((A \cup B)^*)$$

$$\text{由于 } (L^*)^* = L^*$$

$$\text{则有 } L((A^* \cup B^*)^*) \subseteq L(((A \cup B)^*)^*) = L((A \cup B)^*)$$

$$\text{综上有 } (A \cup B)^* = (A^* \cup B^*)^*$$

b. ~~对 $\forall E \in L^*$ 都有~~

$$\text{因为 } E \in L(B^*)$$

$$\text{则 } L(A) = L(A \cdot \{E\}) \subseteq L(A^* \cdot B^*)$$

$$\text{同理 } L(B) \subseteq L(A^* \cdot B^*)$$

$$\text{因此 } L(A \cup B) \subseteq L(A^* \cdot B^*)$$

$$\text{由单调性可知 } L((A \cup B)^*) \subseteq L((A^* \cdot B^*)^*)$$

$$\text{又有 } L(A^* \cdot B^*) \subseteq L((A \cup B)^* \cdot (A \cup B)^*) = L((A \cup B)^*)$$

$$\text{所以 } L((A^* \cdot B^*)^*) \subseteq L(((A \cup B)^*)^*) = L((A \cup B)^*)$$

$$\text{综上有 } (A \cup B)^* = (A^* \cdot B^*)^*$$

8. (1) $(0|1)^*01$

(2) $(0111213141516171819)^*(015)$

12. (a). 起始状态: $A = \{0\}$

① 从 A 出发, $\delta(0, a) = \{0, 1\}$, 记为 B

$\delta(0, b) = \{1\}$, 记为 C

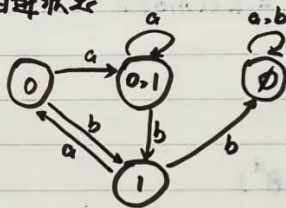
② 从 B 出发, $\delta(0, a) \cup \delta(1, a) = \{0, 1\} = B$

$\delta(0, b) \cup \delta(1, b) = \{1\} = C$

③ 从 C 出发, $\delta(1, a) = A$

$\delta(1, b) = \emptyset$, 记为 D

④ D 为陷阱状态



(b). 终态组 $G_1 = \{0, 1\}$

$S_0 = \{0, 1\}$ $S_1 = \{2, 4\}$ $S_2 = \{3, 5\}$

非终态组 $G_2 = \{2, 3, 4, 5\}$

① $a: 0 \rightarrow 1 \in G_1$ $1 \rightarrow 1 \in G_1$

$b: 0 \rightarrow 2 \in G_2$ $1 \rightarrow 4 \in G_2$

G_1 不可分.

② $a: 2 \rightarrow 0 \in G_1$ $3 \rightarrow 3 \in G_2$

$4 \rightarrow 1 \in G_1$ $5 \rightarrow 5 \in G_2$

G_2 可划分为 $\{2, 4\}$ 和 $\{3, 5\}$

